4(a)(i)

$$F_E = F_B$$

$$qE = Bqv$$

$$v = \frac{E}{B}$$

Centripetal force is provided by magnetic force

$$\frac{mv^2}{r} = Bqv$$

$$r = \frac{mv}{qB}$$

$$= \frac{m}{qB}\frac{E}{B}$$

$$r = \frac{mE}{qB^2}$$

(ii) For ions to have a higher speed, values of *E* and *B* are selected to obtain a larger ratio $\frac{E}{B}$ either by an increase in E or a decrease in B

(iii) For negative charge to turn anti clockwise, magnetic field must now be pointing out of the page. For the ion to continue to move in a straight line in the velocity selector, the electric field must point from right to left

(b)(i) ^{13}C ions have the larger radius

(ii) $$distance\ apart = 2(0.203 - 0.187) = 0.032\ m$$